

Adaptive multi-mesh FEM for the spectral fractional Laplacian

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Motivation & definition

Motivation & definition

$$(-\Delta)^s u = f \quad \text{in } \Omega, \quad \text{with } 0 < s < 1.$$

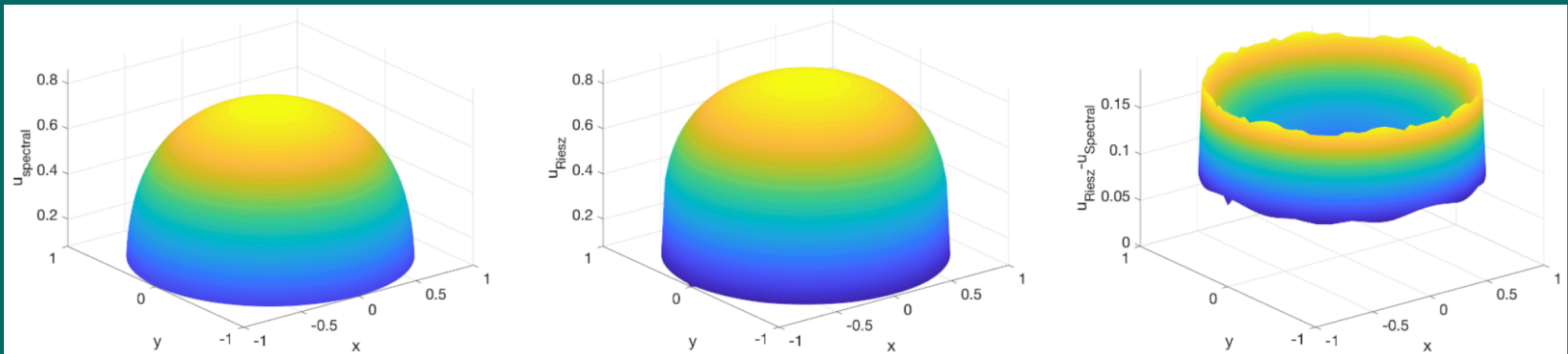
Fractional PDEs arise in various contexts:

- to model non-local behavior
(e.g. diffusion in porous medium¹, surface quasi-geostrophic flow²),
- to generate (Gaussian) random fields
(e.g. spatial statistics³),
- in domain decomposition preconditioning⁴,
- to reconstruct poorly regular functions
(e.g. to solve PDEs “without” BCs⁵).

Motivation & definition

$$(-\Delta)^s u = f \quad \text{in } \Omega, \quad \text{with } 0 < s < 1.$$

The fractional Laplacian admits several non-equivalent definitions on a bounded domain¹.



Motivation & definition

We are interested in a particular definition: the spectral fractional Laplacian.

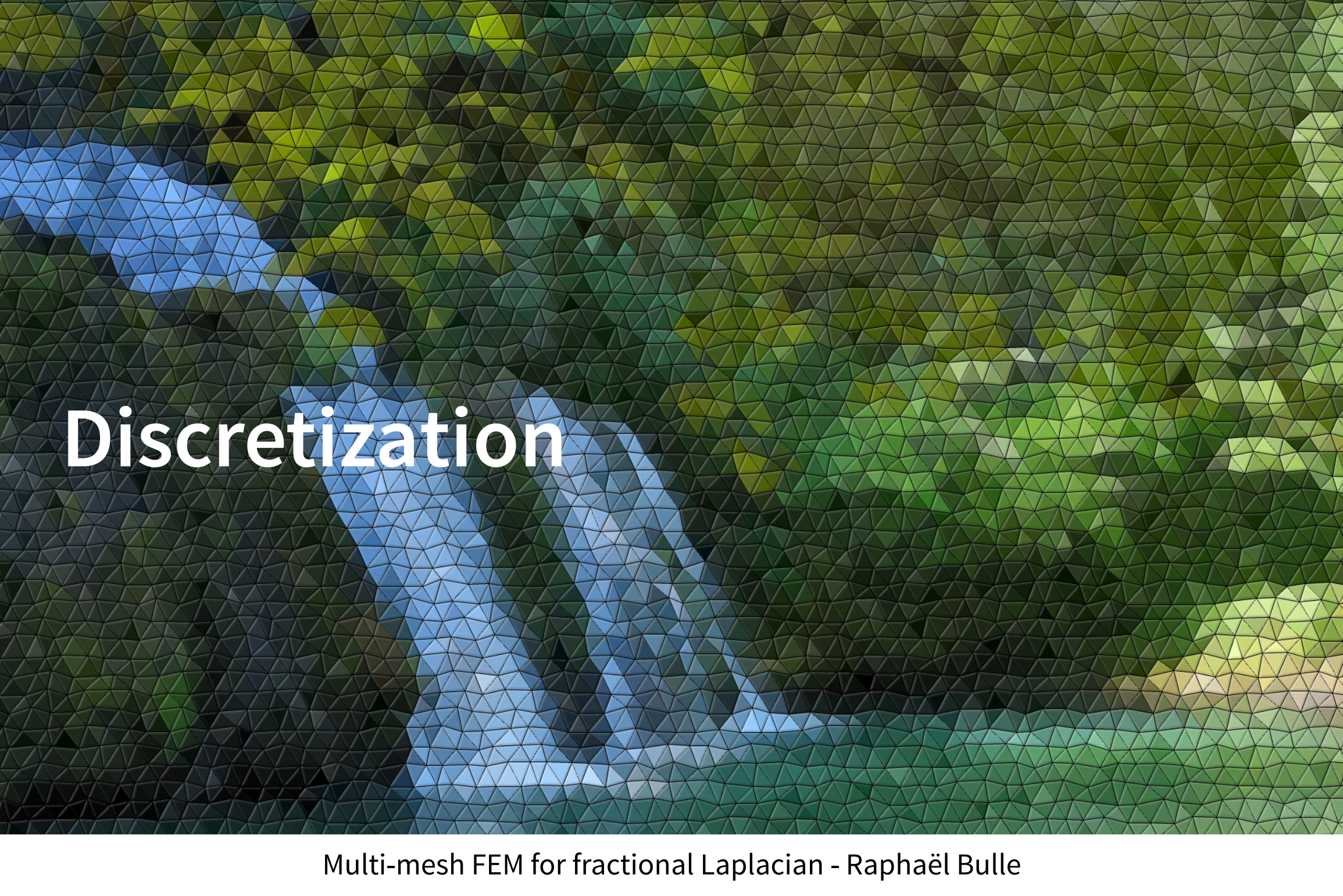
- $s \in (0, 1)$,
- $\Omega \subset \mathbb{R}^2$ a bounded domain with polygonal boundary $\partial\Omega$,
- $f \in L^2(\Omega)$,
- $\{(\psi_i, \lambda_i)\}_{i=1}^{+\infty} \subset L^2(\Omega) \times \mathbb{R}^{>0}$
the spectrum of $-\Delta$ on Ω with homogeneous zero DBC.

Consider:

$$(-\Delta)^s u = f \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega, \quad (1)$$

where the fractional power is taken in the spectral sense:

$$u = \sum_{i=1}^{+\infty} \lambda_i^{-s} (f, \psi_i)_2 \psi_i.$$



Discretization

Discretization

How to solve (1) numerically ?

- Extension to semi-infinite cylinder ¹:

Dirichlet-to-Neumann map to reformulate into a non-fractional problem on an extruded domain.

⚠ 3D domain $\rightarrow$ 4D domain!

- Spectral Element Method²:

Relies on spectral FEM.

⚠ Domain shape.

- Method of heat-semigroup³:

Heat-semigroup to reformulate into a parabolic non-fractional problem.

⚠ Sequence of problems to solve.

- Rational scheme⁴.

Discretization

The rational scheme method is based on the following approximation:

$$Q_s^\kappa(\lambda) := C \sum_{l=1}^{N_\kappa} a_l (c_l + b_l \lambda)^{-1}, \quad \forall \lambda \geq \lambda_0 > 0.$$

Theorem (e.g. Bonito and Pasciak 2015):

One can choose¹ $C \in \mathbb{R}^{>0}$, $\{a_l, b_l, c_l \in \mathbb{R}^{>0}; l = 1, \dots, N_\kappa\}$ s.t.

$$|\lambda^{-s} - Q_s^\kappa(\lambda)| \leq \varepsilon_s(\kappa), \quad \forall \lambda \geq \lambda_0 > 0,$$

with $\varepsilon_s(\kappa) \rightarrow 0$ exponentially fast as $\kappa \rightarrow 0$.

Discretization

Replacing λ_i^{-s} with $Q_s^\kappa(\lambda_i)$ in u we obtain

$$\begin{aligned} u &= \sum_{i=1}^{+\infty} \lambda_i^{-s} (f, \psi_i)_2 \psi_i \approx \sum_{i=1}^{+\infty} Q_s^\kappa(\lambda_i) (f, \psi_i)_2 \psi_i \approx \sum_{i=1}^{+\infty} C \sum_{l=1}^{N_\kappa} a_l (c_l + b_l \lambda_i)^{-1} (f, \psi_i)_2 \psi_i \\ &\approx C \sum_{l=1}^{N_\kappa} a_l \sum_{i=1}^{+\infty} (c_l + b_l \lambda_i)^{-1} (f, \psi_i)_2 \psi_i \\ &\approx C \sum_{l=1}^{N_\kappa} a_l u_l := u_\kappa. \end{aligned}$$

The functions $(u_l)_{l=1}^{N_\kappa}$ are solutions to

$$c_l u_l - b_l \Delta u_l = f \quad \text{in } \Omega, \quad u_l = 0 \quad \text{on } \partial\Omega \quad \forall l = 1, \dots, N_\kappa.$$

Discretization

To sum up:

$$(-\Delta)^s u = f \quad \longleftrightarrow \quad \begin{cases} c_1 u_1 - b_1 \Delta u_1 & = f, \\ c_2 u_2 - b_2 \Delta u_2 & = f, \\ \vdots & \\ c_{N_\kappa} u_{N_\kappa} - b_{N_\kappa} \Delta u_{N_\kappa} & = f. \end{cases}$$

$$u \approx u_\kappa = C \sum_{l=1}^{N_\kappa} a_l u_l.$$

However, the approximation u_κ is not computable as is.

Discretization

We add FEM to the mix:

- Let $\mathcal{T}$ be a triangulation of Ω ,
- $V_0^1 \subset H_0^1(\Omega)$ be a (linear) Lagrange space,
- $\{u_{l,1} \in V_0^1\}_{l=1}^{N_\kappa}$ be the solutions to

$$c_l \int_{\Omega} u_{l,1} v_1 + b_l \int_{\Omega} \nabla u_{l,1} \cdot \nabla v_1 = \int_{\Omega} f v_1 \quad \forall v_1 \in V_0^1 \quad \forall l = 1, \dots, N_\kappa,$$

and get a fully computable approximation:

$$u \approx u_\kappa \approx u_{\kappa,1} = C \sum_{l=1}^{N_\kappa} a_l u_{l,1}.$$

Discretization

This method has several practical advantages:

- Easily parallelizable: fully independent parametric problems.
- “Standard” FE machinery only: e.g. linear Lagrange FEM.
- Well-suited to 3D problems: no dense linear systems, no extension to 4D.
- Domain Ω of “any” shape.



Error analysis

Error analysis

$$u \underset{\text{Rational}}{\approx} u_\kappa \underset{\text{FEM}}{\approx} u_{\kappa,1}$$

- Rational scheme error: $\|u - u_\kappa\|_{L^2(\Omega)}$,
- FEM error: $\|u_\kappa - u_{\kappa,1}\|_{L^2(\Omega)}$.

Theorem (Bonito and Pasciak 2015)

Let u be solution to (1) and u_κ be its rational approximation. For any $\kappa > 0$ we have

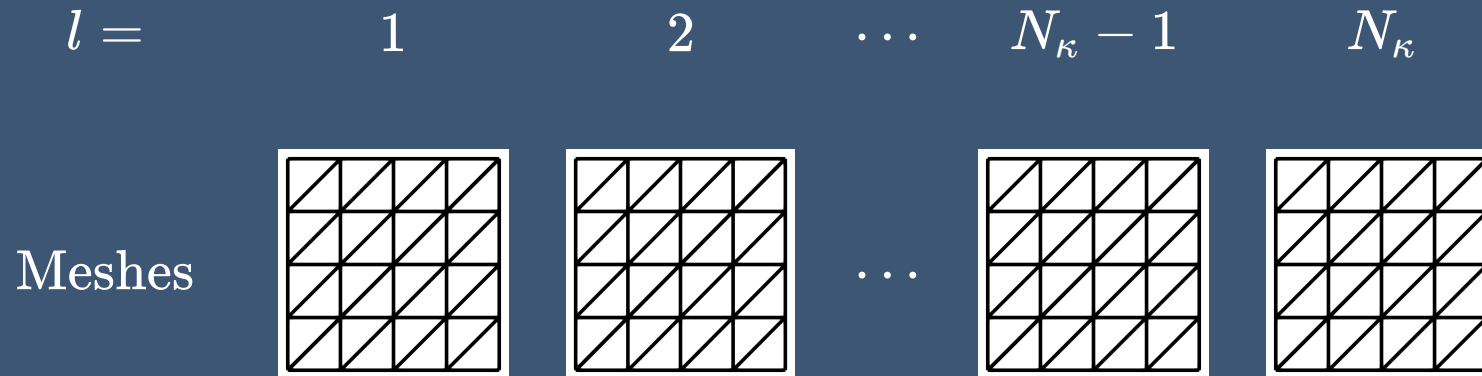
$$\|u - u_\kappa\|_{L^2(\Omega)} \leq \varepsilon_s(\kappa) \|f\|_{L^2(\Omega)},$$

where ε_κ is **known** and converges to 0 as $\kappa \rightarrow 0$ exponentially fast.

Therefore, we neglect the rational scheme error and consider

$$\|u - u_{\kappa,1}\|_{L^2(\Omega)} = \|u_\kappa - u_{\kappa,1}\|_{L^2(\Omega)}.$$

Error analysis

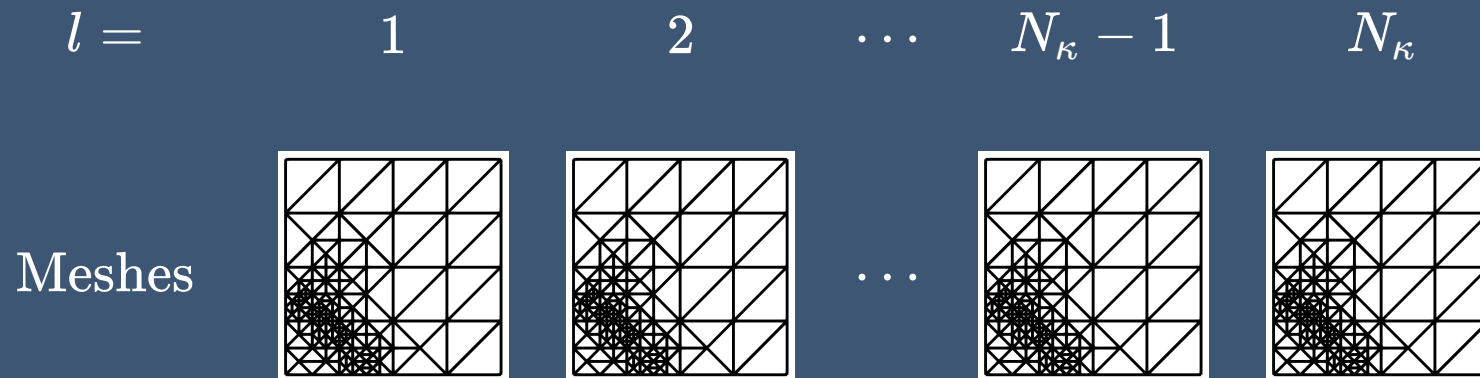


In Bulle et al. (2023) we introduce a FEM error estimator based on the **Bank–Weiser estimator**¹:

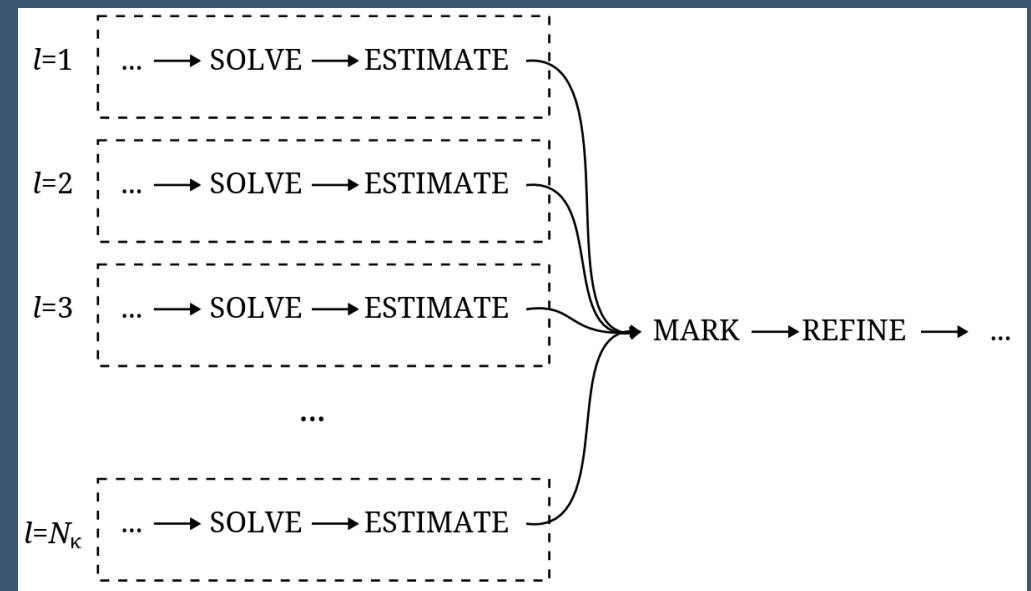
$$\|u_\kappa - u_{\kappa,1}\|_{L^2(T)}^2 = \left\| C \sum_{l=1}^{N_\kappa} a_l (u_l - u_{l,1}) \right\|_{L^2(T)}^2 \approx^2 \left\| C \sum_{l=1}^{N_\kappa} a_l e_{l,T} \right\|_{L^2(T)}^2 ,$$

for each l and T , $e_{l,T} \approx^2 (u_l - u_{l,1})|_T$ are solutions to **problems localized to each cell T** .

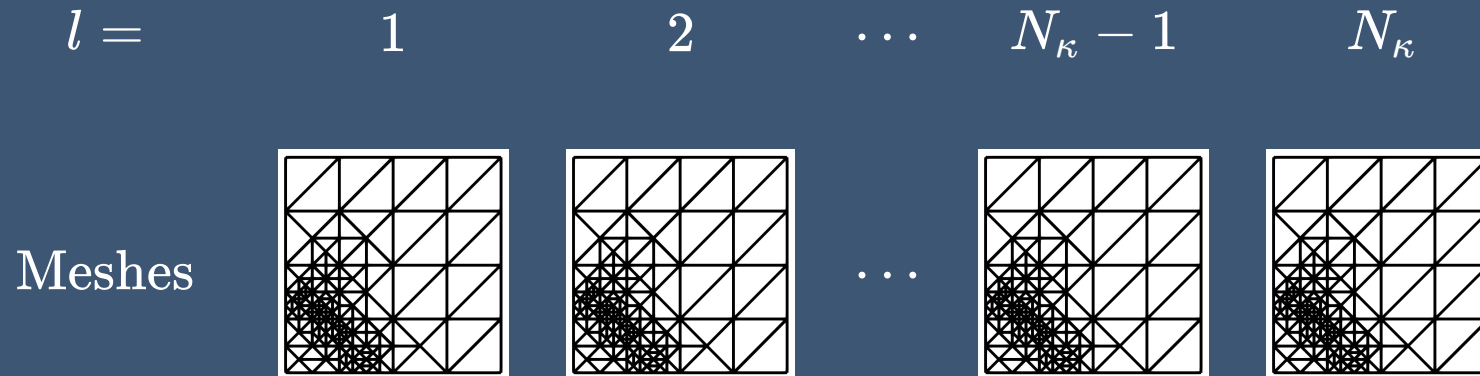
Error analysis



Using this estimator, we perform adaptive mesh refinement.



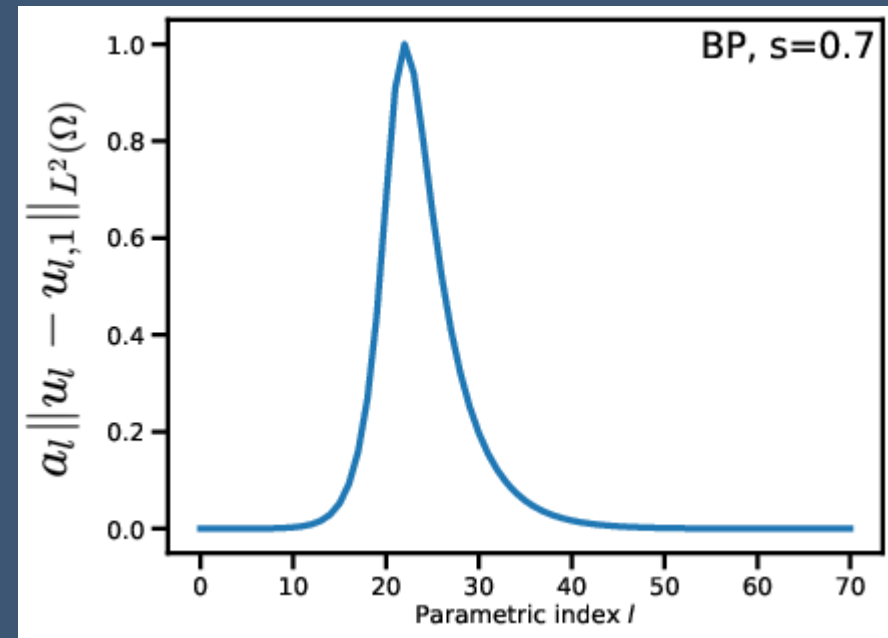
Error analysis



Using this estimator, we perform adaptive mesh refinement.

However, this approach has a limitation:

FE errors are not uniform w.r.t. l





Multi-mesh framework

Multi-mesh FEM for fractional Laplacian - Raphaël Bulle

Multi-mesh framework

Since,

$$\|u_\kappa - u_{\kappa,1}\|_{L^2(\Omega)} \leqslant C \sum_{l=1}^{N_\kappa} a_l \|u_l - u_{l,1}\|_{L^2(\Omega)}.$$

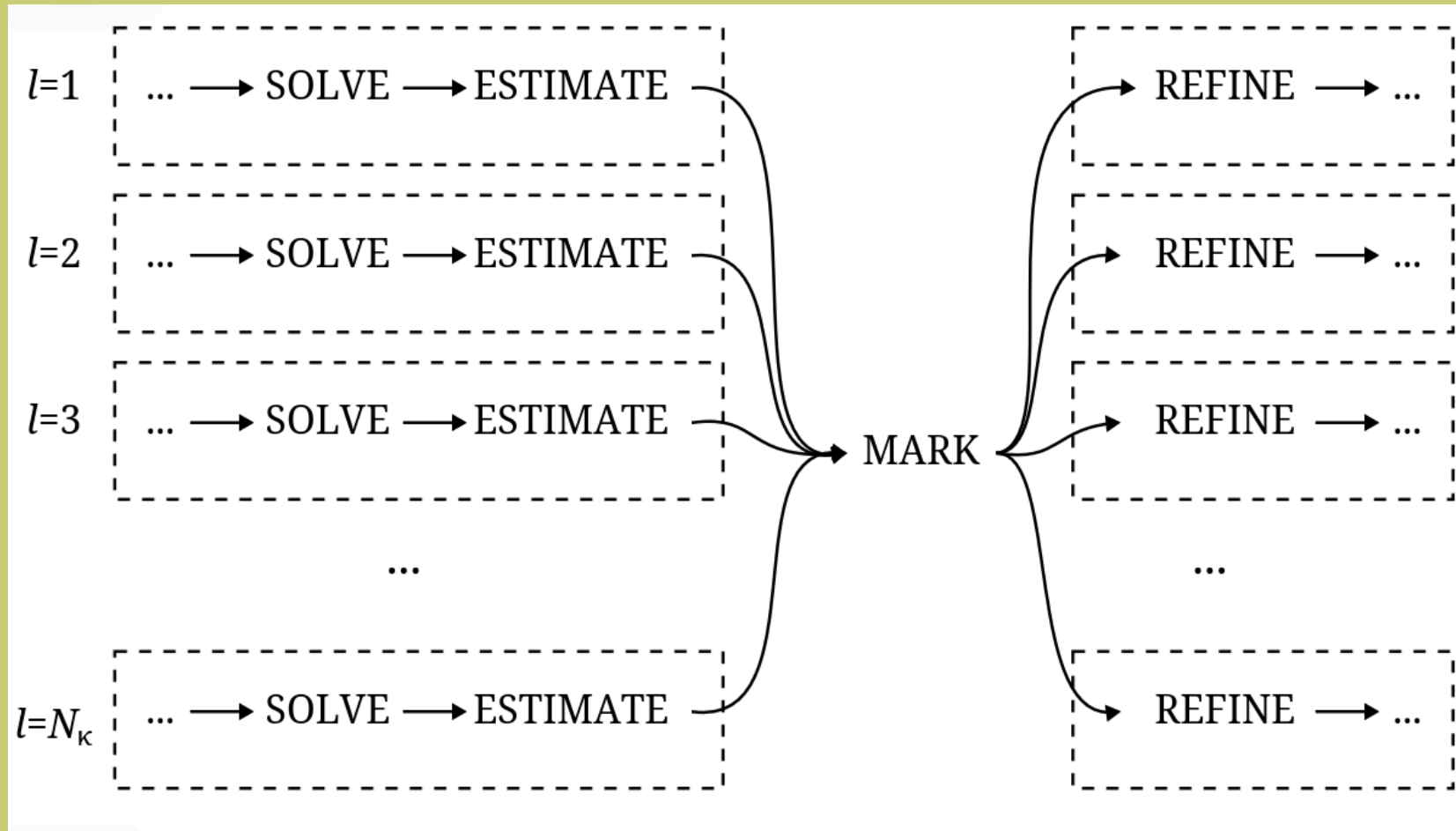
it is sufficient to have

$$a_l \|u_l - u_{l,1}\|_{L^2(\Omega)} \leqslant \frac{\text{tol}}{CN_\kappa} \quad \forall l = 1, \dots, N_\kappa.$$

Proposed solution from Bespalov and Bulle (2026):

Use a different mesh for each l and refine these meshes adaptively “w.r.t. l ”.

Multi-mesh framework



Bespalov and Bulle (2026) multi-mesh algorithm outline.

Multi-mesh framework

SOLVE

Let $\mathcal{T}_l$ for $l = 1, \dots, N_\kappa$ on Ω and $V_{l,0}^1 \subset H_0^1(\Omega)$ the corresponding linear Lagrange FE spaces.

As previously, we solve

$$c_l \int_{\Omega} u_{l,1} v_{l,1} + b_l \int_{\Omega} \nabla u_{l,1} \cdot \nabla v_{l,1} = \int_{\Omega} f v_{l,1} \quad \forall v_{l,1} \in V_{l,0}^1 \quad \forall l = 1, \dots, N_\kappa,$$

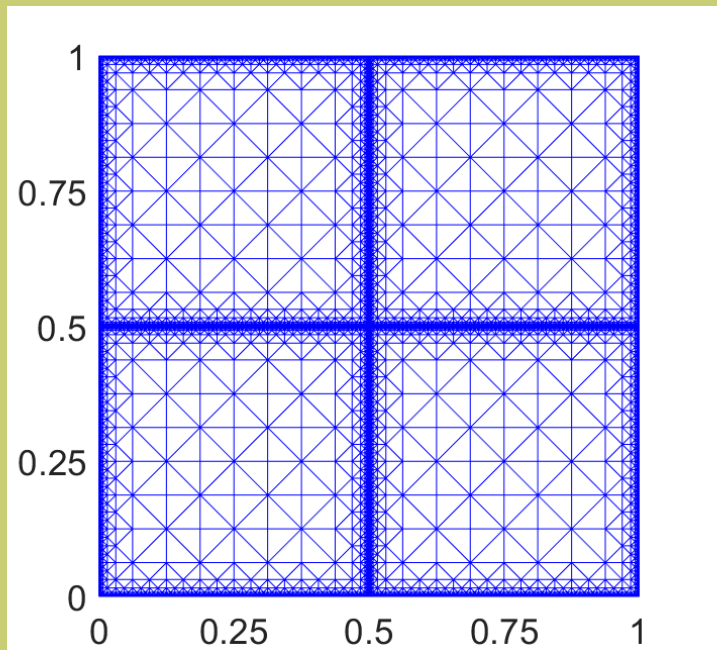
and compute

$$u_{\kappa,1} = C \sum_{l=1}^{N_\kappa} a_l u_{l,1}.$$

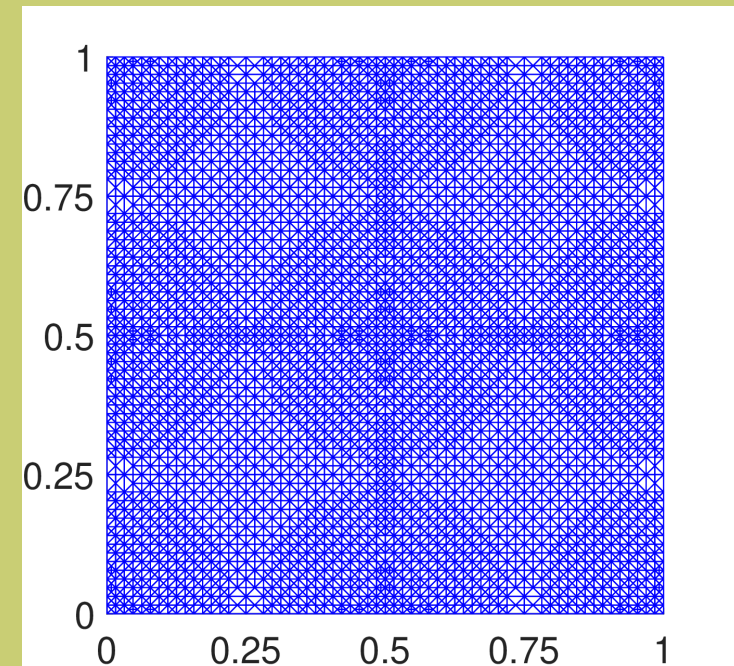
Multi-mesh framework

SOLVE

⚠ The $(u_{l,1})_l$ do not live on the same mesh anymore!



$l = 82$



$l = 123$

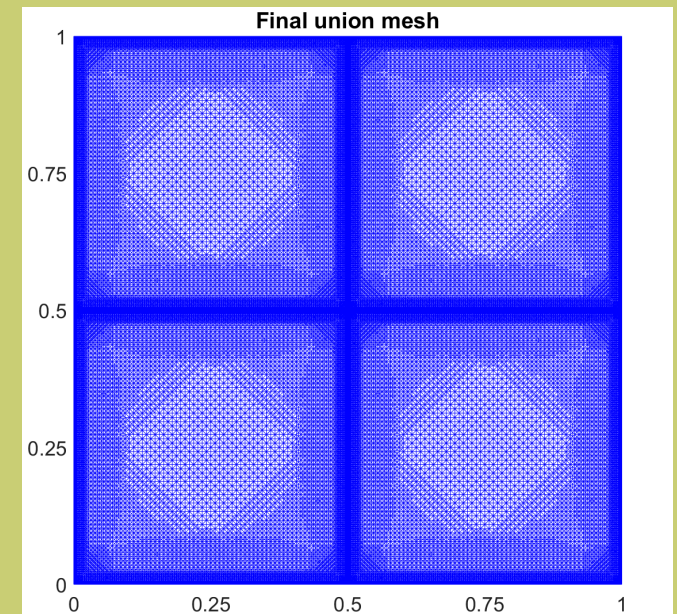
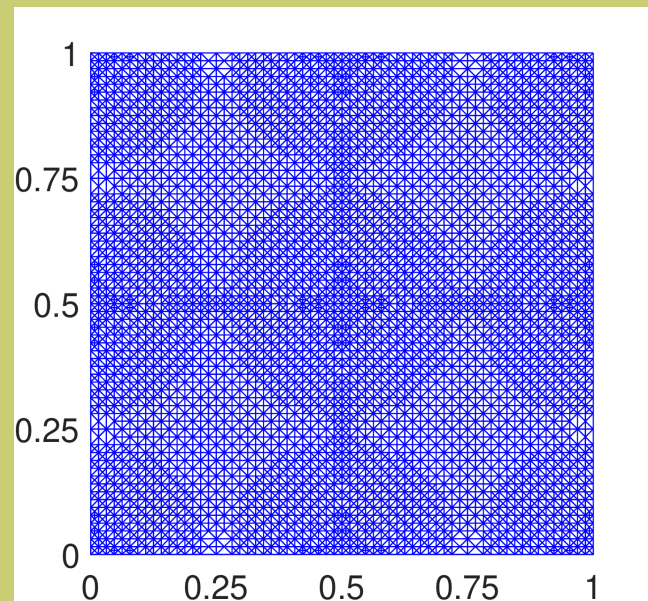
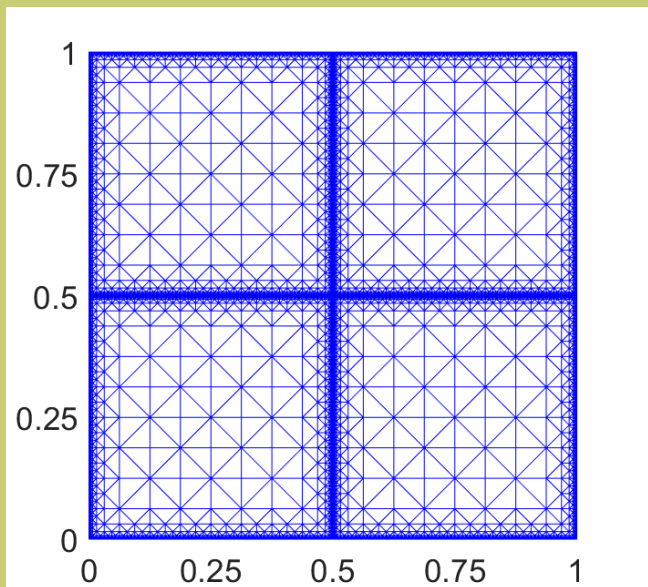
Multi-mesh framework

SOLVE

⚠ The $(u_{l,1})_l$ do not live on the same mesh anymore!

In practice, we need a union mesh $\tilde{\mathcal{T}}$ to interpolate the $(u_{l,1})_l$ before the sum.

We compute $\tilde{\mathcal{T}}$ using a constrained Delaunay triangulation algorithm¹.



Multi-mesh framework

ESTIMATE

- Local error estimation: ✓ same as before
we compute $e_{l,T_l} \approx (u_l - u_{l,1})|_{T_l}$ for $T_l \in \mathcal{T}_l$ and $l = 1, \dots, N_\kappa$ using Bank-Weiser strategy.
- Global error estimation: ⚠ we can not sum the functions e_{l,T_l} for $l = 1, \dots, N_\kappa$ anymore since they do not live on the same cell!

Single mesh case reminder:

$$\left[\sum_{T \in \mathcal{T}} \left\| C \sum_{l=1}^{N_\kappa} a_l e_{l,T} \right\|_{L^2(T)}^2 \right]^{1/2}$$

Multi-mesh framework

ESTIMATE

We try to approaches

Triangle inequality based error estimation:

$$\begin{aligned} \|u_\kappa - u_{\kappa,h}\|_{L^2(\Omega)} &\leq C \sum_{l=1}^{N_\kappa} a_l \|u_l - u_{l,1}\|_{L^2(\Omega)} = C \sum_{l=1}^{N_\kappa} a_l \left[\sum_{T_l \in \mathcal{T}_l} \|u_l - u_{l,1}\|_{L^2(T_l)}^2 \right]^{1/2} \\ &\approx C \sum_{l=1}^{N_\kappa} a_l \left[\sum_{T_l \in \mathcal{T}_l} \|e_{l,T_l}\|_{L^2(T_l)}^2 \right]^{1/2} := \eta \end{aligned}$$

“First spatial sum then rational sum.”

Multi-mesh framework

ESTIMATE

We try to approaches

Union mesh based error estimation:

by interpolating the solutions $(u_{l,1})_l$ onto $\tilde{\mathcal{T}}$ we can compute $e_{l,\tilde{T}} \approx (u_l - \tilde{u}_{l,1})|_{\tilde{T}}$ for each l and $\tilde{T}$ and obtain, as in the single mesh case

$$e_{\tilde{T}} := C \sum_{l=1}^{N_\kappa} a_l e_{l,\tilde{T}} \quad \forall \tilde{T} \in \tilde{\mathcal{T}} \quad \text{and} \quad \|u_\kappa - u_{\kappa,1}\|_{L^2(\Omega)}^2 \approx \sum_{\tilde{T} \in \tilde{\mathcal{T}}} \|e_{\tilde{T}}\|_{L^2(\tilde{T})}^2 =: \tilde{\eta}^2.$$

“First rational sum then spatial sum.”

Multi-mesh framework

MARKING

We adapt Dörfler's marking strategy¹ to the multi-mesh framework:

Let $\mathcal{M} := \bigcup_{l=1}^{N_\kappa} \mathcal{M}_l \subset \bigcup_{l=1}^{N_\kappa} \mathcal{T}_l$. Given $\theta \in (0, 1]$ we seek $\mathcal{M}$ of minimal cardinality such that

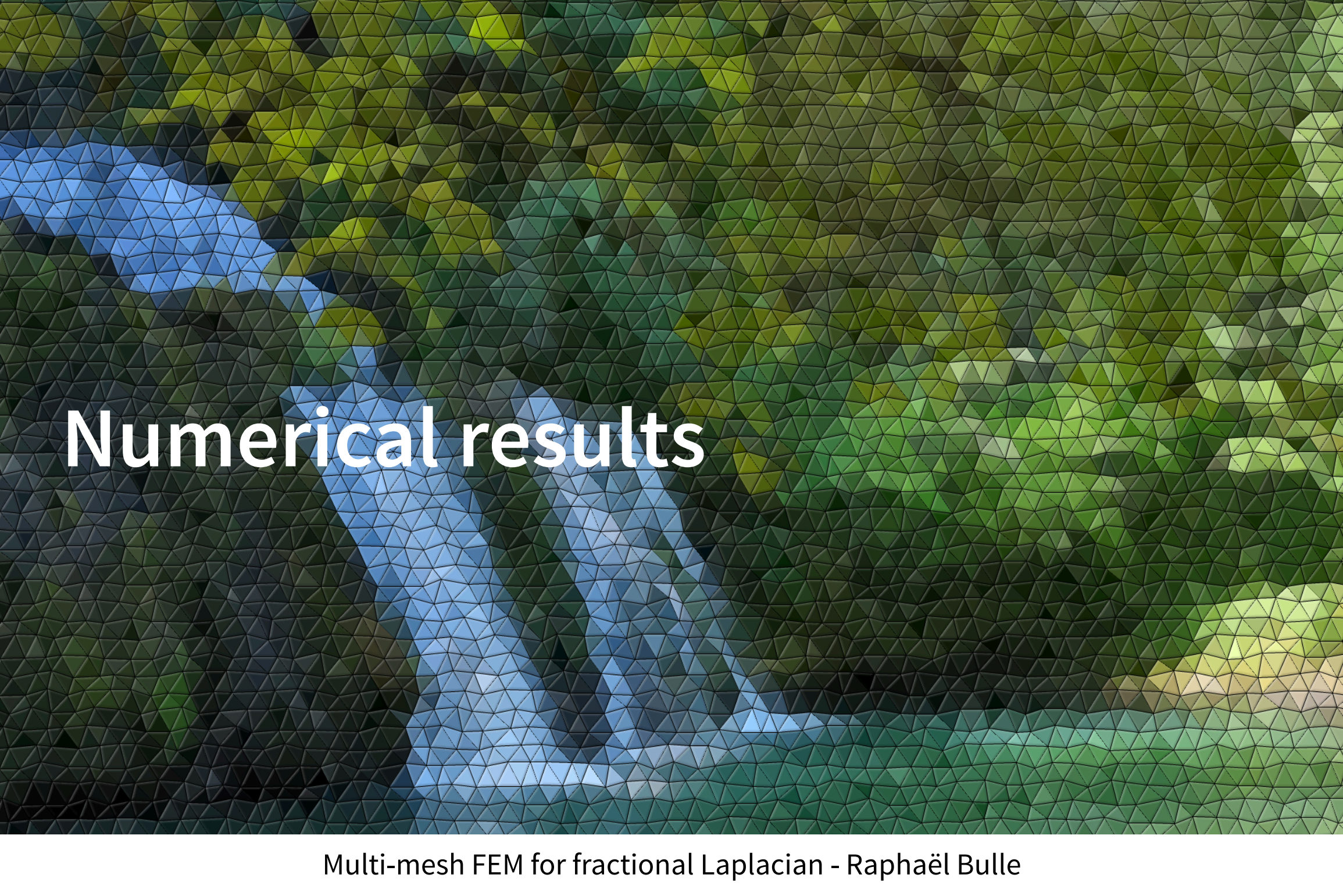
$$\theta \left[\sum_{l=1}^{N_\kappa} \sum_{T \in \mathcal{T}_l} (a_l \eta_{l,T})^2 \right] \leq \sum_{l=1}^{N_\kappa} \sum_{T \in \mathcal{M}_l} (a_l \eta_{l,T})^2.$$

Multi-mesh framework

REFINE

For each l we refine $\mathcal{T}_l$ according to the corresponding $\mathcal{M}_l$.

Note: If any $\mathcal{M}_l = \emptyset$, then the mesh $\mathcal{T}_l$ will not be refined.
We can keep $u_{l,1}$ and $(e_{l,T_l})_{T_l \in \mathcal{T}_l}$ for the next iteration!



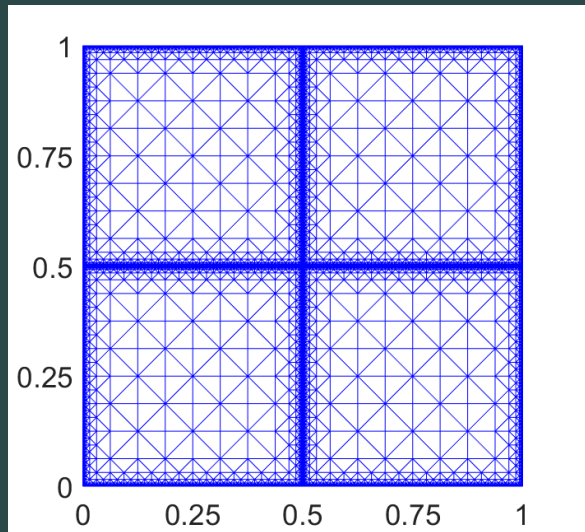
Numerical results

Numerical results

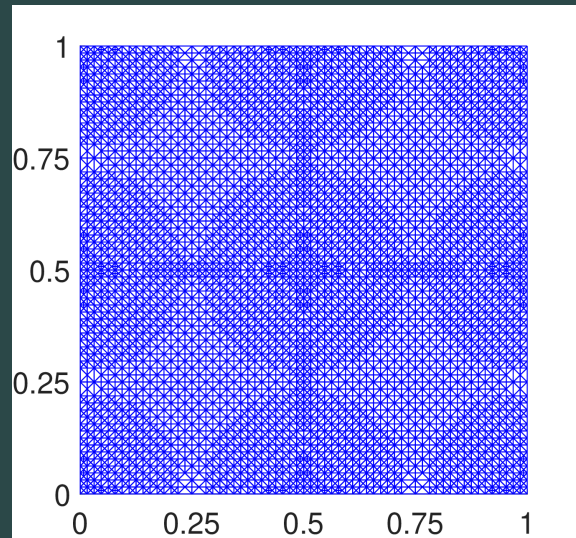
Our method is implemented using T-IFISS package in Matlab¹.

$$\Omega = (0, 1)^2 \text{ and } f(x, y) = \text{sgn}((x - 0.5)(y - 0.5)).$$

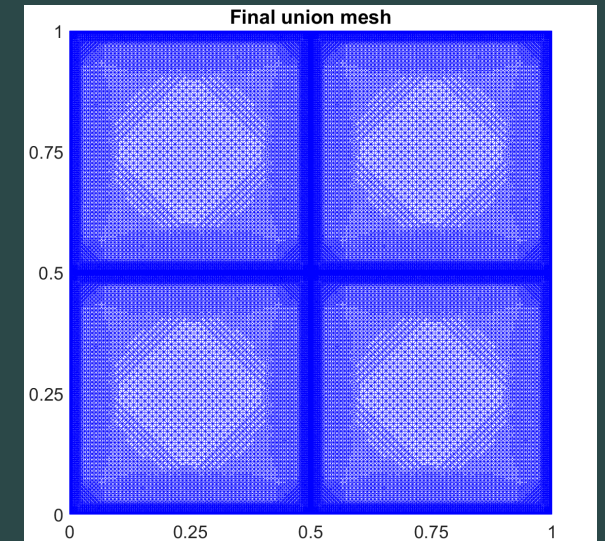
Meshes for $s = 0.3$.



$\mathcal{T}_{82}$



$\mathcal{T}_{123}$

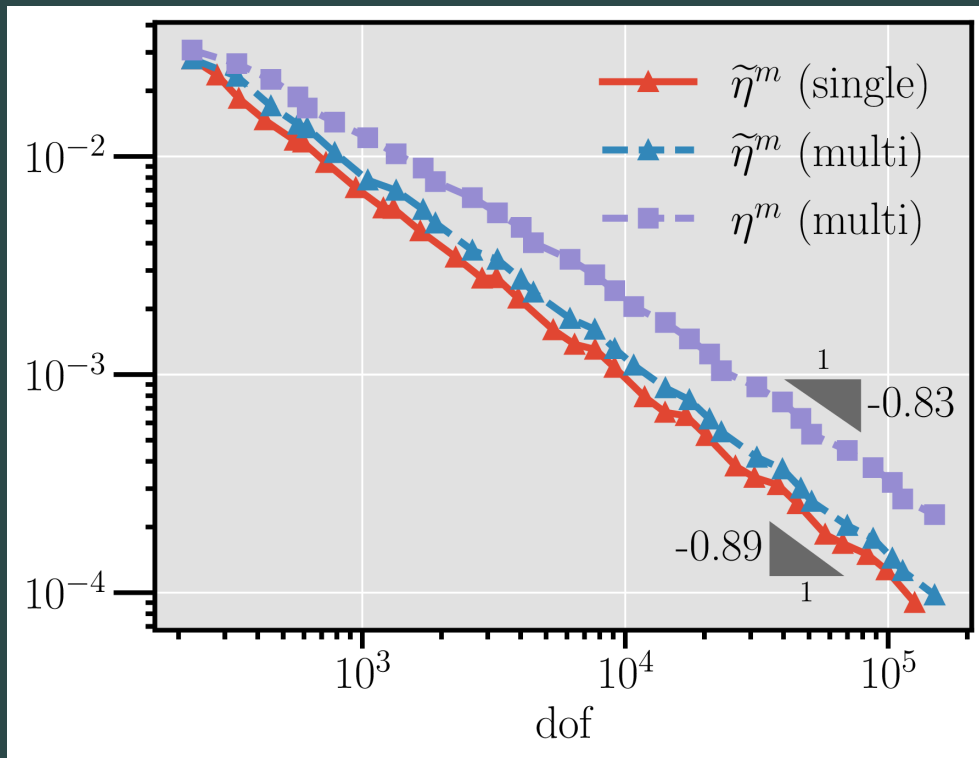


$\tilde{\mathcal{T}}$

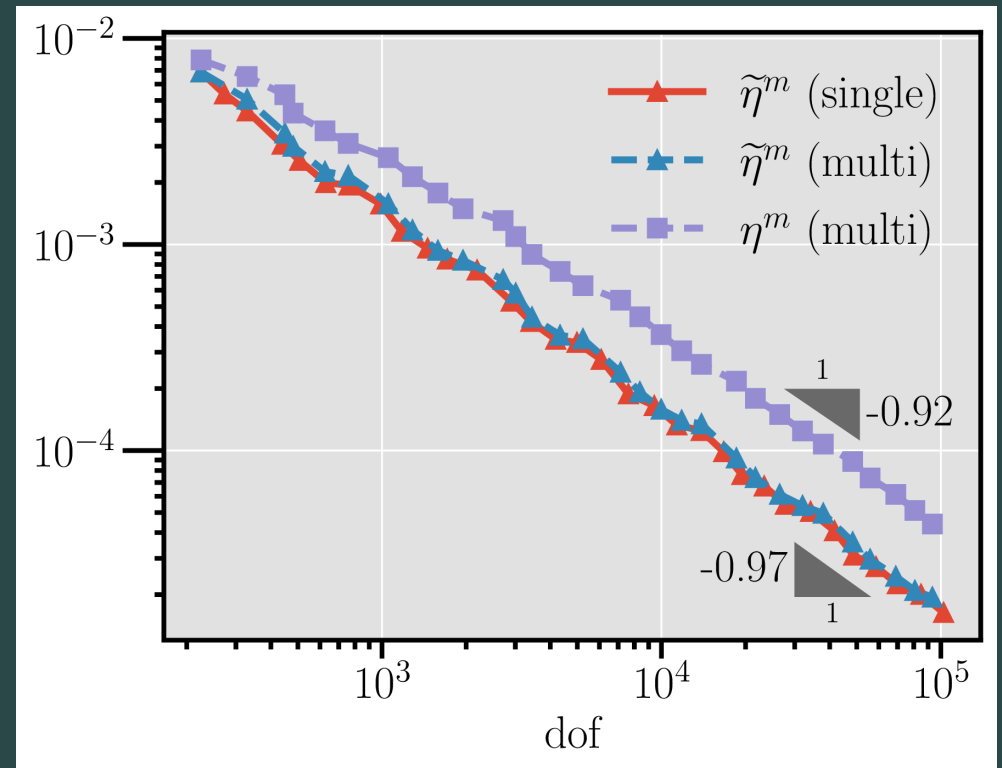
Numerical results

Convergence curves

$\tilde{\eta}$: union mesh global estimator ; η : triangle inequality global estimator.



$s = 0.3$



$s = 0.5$

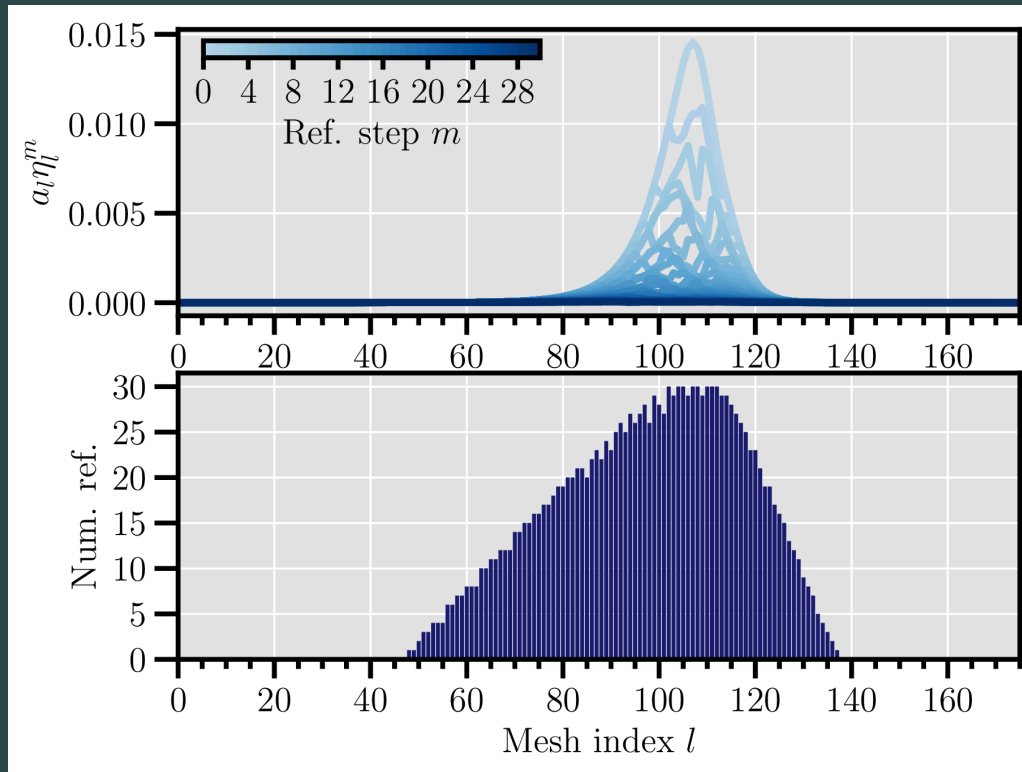
Numerical results

Convergence rates

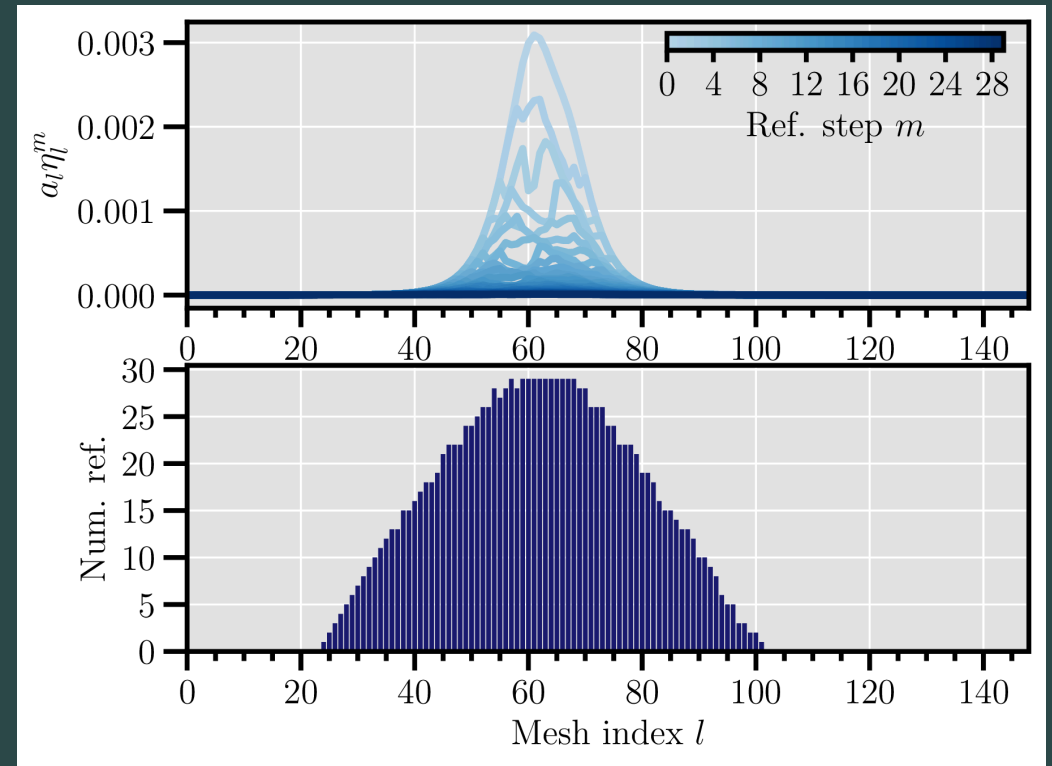
s	0.1	0.3	0.5	0.7	0.9
Theory ¹ (uniform)	-0.35	-0.55	-0.75	-0.95	-1.00
$\tilde{\eta}^m$ (single)	-0.67	-0.90	-0.98	-1.00	-0.99
$\tilde{\eta}^m$ (multi)	-0.66	-0.91	-0.99	-1.01	-0.99
η^m (multi)	-0.62	-0.83	-0.92	-0.95	-0.97

Numerical results

Performance



$s = 0.3$

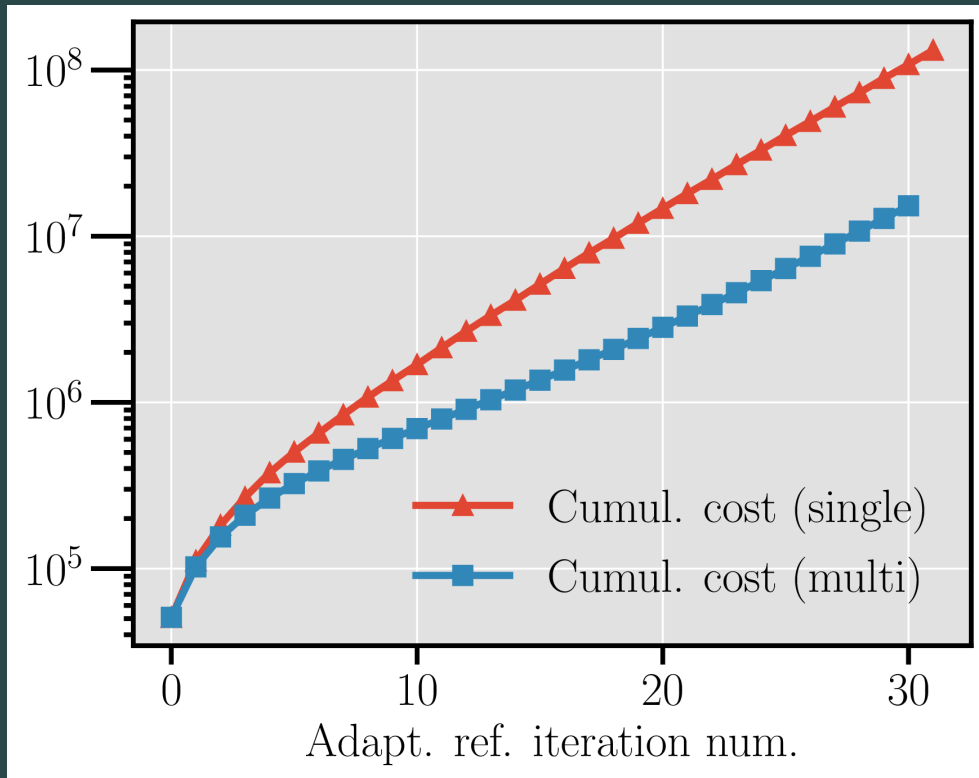


$s = 0.5$

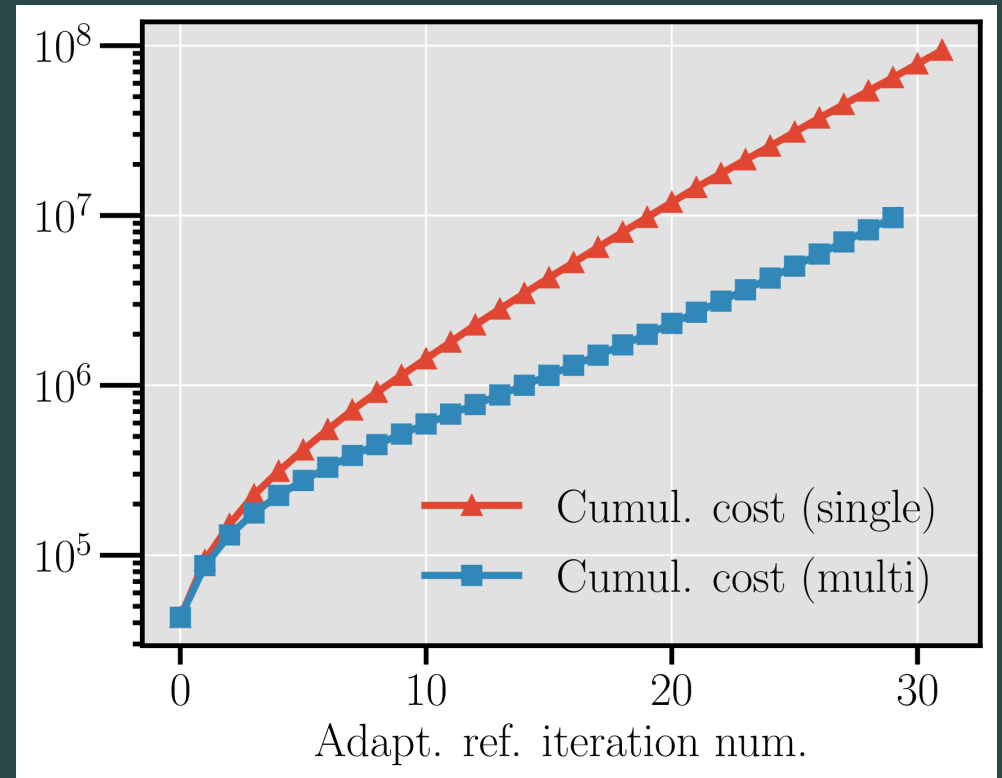
Numerical results

Performance

$$\text{Cumul. cost} = \sum_l \# \text{dof.}$$



$s = 0.3$



$s = 0.5$



Conclusion & perspective

Conclusion & perspective

Summary

- We design a new multi-mesh adaptive refinement scheme to solve the spectral fractional Laplacian.
- We get satisfactory convergence rates (consistent with the single-mesh approach).
- We get a computational cost divided by ~ 10 on some test cases.

Perspective

- Combine the adaptive rational scheme from Bulle et al. (2023) with our multi-mesh approach (ongoing work).
- Optimize the computation of the union mesh.
- Implement the algorithm in parallel.
- Adapt to other definitions of the fractional Laplacian.
- Derive an error estimator in the fractional norm.
- Maths proofs ??

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Thank you
for your attention!

Any question ?